FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2024
CONCEPT RECAPITULATION TEST – IV
PAPER –1
TEST DATE: 01-05-2024

ANSWERS, HINTS & SOLUTIONS

Physics

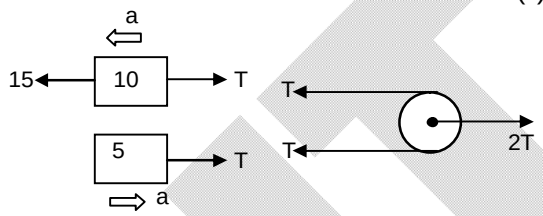
PART – I

SECTION – A

1. ACD

Sol. $15 - T = 10a$ ….(i)

$T = 5a$ ….(ii)



So, $a = 1\ m/s^2$

And $T = 5\ N$

So, $2T = 10\ N$

2. ACD

Sol. $T_A = T_B = \dfrac{(300+400)}{2} = 350\ ^oC$

$\therefore\ T_A = T_B = T_C$

3. ABD

Sol. $\vec{\tau} = \vec{P} \times \vec{E} = \left(0.6\hat{i} - 0.4\hat{j}\right)$

$U = \vec{P} \cdot \vec{E} = -0.6J$

4. C

Sol. The momentum will be same along the ground before and after collision

V_1 before collision perpendicular to ground

$= -25 \times \sin 53^o = -20\ m/s$

V_2 after collision perpendicular to ground

$= 15\sqrt{2} \times \sin 45^o = 15\ m/s$

$\Delta P = mV_2 - mV_1$

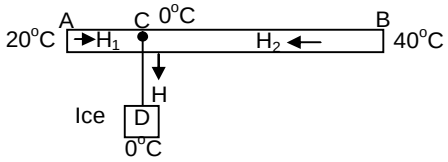
$= 1\times 15 - 1\times(-20)$

$= 35$ kg m/s

Impulse exerted by ground = $\Delta P = 35\ N - sec$.

5. D

Sol.



Thermal resistance of $AC\left(=\frac{L}{KA}\right) = \frac{0.1}{336\times 1\times 10^{-1}}$

$= \frac{10^3}{336} = R(\text{suppose})$

Thermal resistance of BC = $\frac{0.2}{336\times 10^{-4}} = 2R$

Temperature of C = 0^oC

$\therefore H_1 = \frac{20}{R}; H_2 = \frac{40}{2R} = \frac{20}{R}$

$\therefore H = H_1 + H_2 = \frac{40}{R} = \frac{40\times 336}{10^3} = \frac{13440}{10^3}$

$= 13.44$ watt

Rate of melting of ice = $\frac{H}{L_f} = \frac{4.2}{80} g/s = 40\ mg/s$.

6. B

Sol. $\frac{8 - V_P}{4} = \frac{V_P - 3}{1}$

$\Rightarrow V_P = 4$

$U = \frac{1}{2}\times 3\times 4^2\ \mu J$

$= 24\ \mu J$

7. A

Sol. $E = -\frac{d\phi}{dt} = aB_0Ae^{-at}$ (where A is area of the loop)

8. B

Sol. (P) $v = 2t^2$

$a = \frac{dv}{dt} = 4t$

$s = \int v\ dt = \frac{2t^3}{3}$

$w.d = \Delta KE = \frac{1}{2}m\Delta\left(v^2\right)$

$= \frac{1}{2}m\left[2\left(t_2^2 - t_1^2\right)\right]$ and $t_2 > t_1$

w.d. is +ve

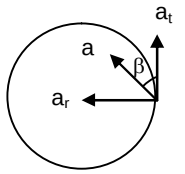
Power of F v

$= mav$

$= m(4t)\,(2t^2)$

$= 8\,mt^3 \neq 0$

$$\tan\beta = \frac{a_r}{a_t} = \frac{\frac{V^2}{R}}{\left(\frac{dv}{dt}\right)}$$



$$= \frac{\left(2t^2\right)^2}{R \cdot 4t} = \frac{t^3}{R} \qquad \text{(Variable)}$$

$$a = \sqrt{a_t^2 + a_r^2} = \sqrt{(4t)^2 + \frac{\left(2t^2\right)^2}{R}} \qquad \text{(Variable)}$$

(Q) $KE = 2s^2$

$$\frac{1}{2}mv^2 = 2S^2$$

$$\frac{1}{2}(1)v^2 = 2S^2$$

$v = 2S$

$$a_t = \frac{vdv}{dS} = 4S$$

$w.d. = \Delta KE$

$= 2\left(S_2^2 - S_1^2\right) > 0$

Power $= \vec{F} \cdot \vec{v}$

$= m\,a_t\,v$

$= (1)\,(4S)\,(2S)$

$= 8S^2 > 0$

$$\tan\beta = \frac{a_r}{a_t} = \frac{\frac{v^2}{R}}{\left(\frac{dv}{dt}\right)} = \frac{1}{4s} = S > 0 \qquad \text{(variable)}$$

$$a = \sqrt{a_t^2 + a_r^2}$$

$$= \sqrt{(4S)^2 + (2S)^4 > 0} \qquad \text{(variable)}$$

(R) $\omega = 2t$

$$\alpha = \frac{d\omega}{dt} = 2$$

$$w.d = \Delta KE = \frac{1}{2}m\left(v_f^2 - \omega_i^2\right)$$

$$= \frac{1}{2}m\left(\omega_f^2 - \omega_i^2\right)R^2$$

$$= \frac{1}{2}m\left[\left(2t_f\right)^2 - \left(2t_i\right)^2\right]$$

$$\text{w.d.} = 2\left(t_f^2 - t_i^2\right) > 0$$

$$\text{Power} = \frac{d(KE)}{dt} = \frac{d}{dt}\left(\frac{1}{2}mv^2\right)$$

$$= m(\omega R)(\alpha R)$$

$$= mR^2(2t)(2)$$

$$= 4t > 0$$

$$\tan\beta = \frac{a_r}{a_t} = \frac{\frac{v^2}{R}}{\left(\frac{dv}{dt}\right)} = \frac{\omega^2 R}{R\alpha} = \frac{(2t)^2}{2}$$

$$= 2t^2 > 0 \qquad \text{(variable)}$$

(S) $a = \sqrt{a_t^2 + a_r^2} = \sqrt{(R\alpha)^2 + \left(R\omega^2\right)^2}$

$$= \sqrt{2^2 + (2t)^4 > 0} \qquad \text{(variable)}$$

$$\theta = 2 + 3t$$

$$\omega = \frac{d\theta}{dt} = 3 \qquad \text{(constant)}$$

$$\alpha = \frac{d\omega}{dt} = 0 \qquad \text{(Uniform circular motion)}$$

$$\text{w.d.} = \Delta KE = 0$$

$$\text{Power} = \frac{d}{dt}(KE) = 0$$

$$\tan\beta = \frac{a_r}{a_t} = \frac{\omega^2 R}{\alpha R} = \infty$$

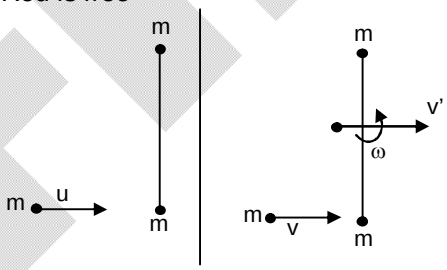
So, $\beta = 90^o$ (constant)

$$a = \sqrt{a_t^2 + a_r^2} = a_r$$

$$= \omega^2 R = 9\ m/s^2$$

9. A

Sol. (P) Rod is free



Conservation of linear momentum:

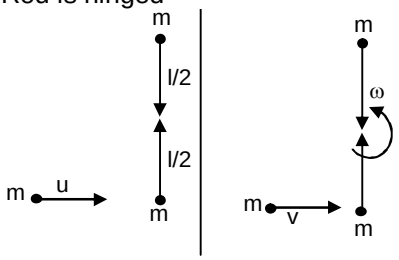
$$mu = mv + 2mv' \qquad \ldots(i)$$

Conservation of angular momentum:

$$mu\frac{l}{2} = mv\frac{l}{2} + \left(2m\frac{l^2}{4}\right)\omega \qquad \ldots(ii)$$

$$eu = \left(v' + \omega\frac{l}{2}\right) - v \qquad \text{(e = coefficient of restitution)} \quad \text{(iii)}$$

(Q) Rod is hinged



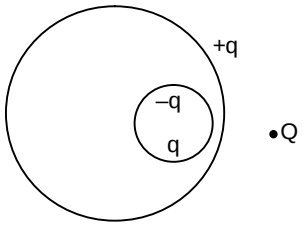
Conservation of angular momentum about hinge

$$mu\frac{l}{2} = mv\frac{l}{2} + 2\left[m\left(\frac{l}{2}\right)^2\right]\omega \qquad \ldots(i)$$

$$eu = \omega\frac{l}{2} - v \qquad \text{(e = coefficient of restitution)} \ldots(ii)$$

10. B

Sol.



(P) If outside charge is shifted to other position, distribution of +q on outer surface changes. Inside cavity charges is unaffected.

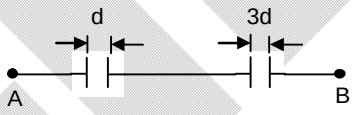
(Q) If inside charges is shifted, (-q) on inner surface will change so that net effect of (+q) and (-q) inside cavity becomes zero outside cavity.

(R) When magnitude of charge inside cavity is changed (-q) inside cavity and (+q) outside cavity also change.

(S) When conductor is earthed charge on outer surface changes so that potential of conductors becomes zero.

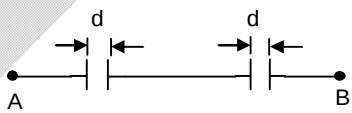
11. D

Sol. (P)



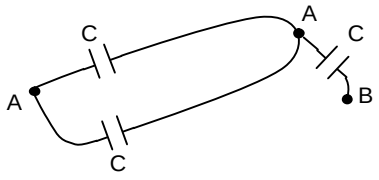
$$\Rightarrow C_{eq} = \frac{C\cdot\frac{C}{3}}{C+\frac{C}{3}} = \frac{C}{4} = \frac{\varepsilon_0 S}{4d}$$

(Q)



$$\Rightarrow C_{eq} = \frac{C}{2} \Rightarrow C_{eq} = \frac{\varepsilon_0 S}{2d}$$

(R)



$$\Rightarrow C_{eq} = \frac{2C}{3} \Rightarrow C_{eq} = \frac{2}{3}\frac{\varepsilon_0 S}{d}$$

(S) $$C_{eq} = \frac{\varepsilon_0\left(\frac{S}{2}\right)}{d} + \frac{\varepsilon_0\left(\frac{S}{2}\right)}{2d} = \frac{3\varepsilon_0 S}{4d}$$

SECTION – B

12. 8

Sol. $$F_{min} = \frac{\mu mg}{\sqrt{1+\mu^2}} \text{ when } \theta = \tan^{-1}(\mu)$$

$$\Rightarrow W = (F_{min})S\cos\theta = \frac{\mu mgS}{(1+\mu^2)} = 8 \text{ joule}$$

13. 80

Sol. $$W_{ext} = q(V_f - V_i)$$

$$e\left[\frac{Kp\cos\theta_f}{r^2} - \frac{Kp\cos\theta_1}{r^2}\right] = W$$

$$\Rightarrow \ -2e\frac{Kp}{r^2} = W$$

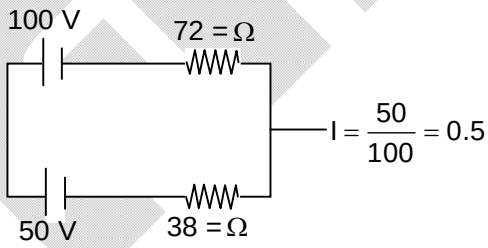
$$\Rightarrow p = -\frac{Wr^2}{2eK}$$

$$= -\frac{\left[\left(-9\times10^{-30}\right)\times\left(16\times10^{-9}\right)^2\right]}{2\times1.6\times10^{-19}\times9\times10^{9}}$$

$$= 80\times10^{-38} \text{ C-m.}$$

14. 80

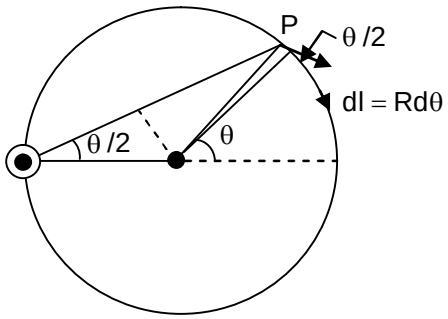
Sol.



Now $$100 - 72\times\frac{1}{2} = \ell_1 \times \frac{100}{1000} \times \frac{800}{100} \Rightarrow \ell_1 = 80 \text{ cm}$$

15. 8

Sol.



$$\text{B at point } P = \frac{\mu_0 i}{2\pi(2R)\cos\frac{\theta}{2}}$$

$$\therefore \quad B \cdot dl = \frac{\mu_0 i}{2\pi(2R)} \frac{\cos\left(\frac{\theta}{2}\right) R d\theta}{\cos\left(\frac{\theta}{2}\right)} = \frac{\mu_0 i d\theta}{4\pi}$$

$$\therefore \int_{DEF} B \cdot dl = \int_0^{\pi/2} \frac{\mu_0 i}{4\pi} d\theta = \frac{\mu_0 i}{8}$$

16. 0

Sol. $I_L = \frac{100}{50\sqrt{2}} = \sqrt{2}A$

$I_C = \frac{100}{50\sqrt{2}} = \sqrt{2}$

$I_{source} = I_L - I_C = 0$

17. 4

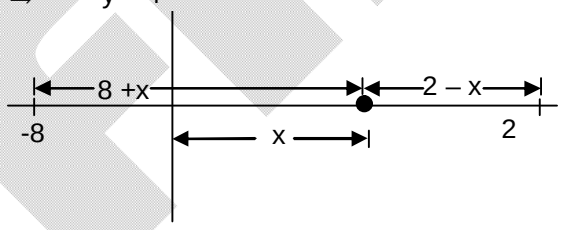
Sol. $t = \frac{8+x}{c} = \frac{2-x}{c}$

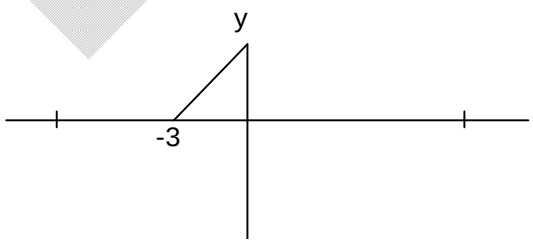
$\Rightarrow \quad -6 = 2x$

$x = -3$ m

$\Rightarrow \quad t = \frac{5}{c} = 5\sqrt{3^2 + y^2}$

$\Rightarrow \quad y = 4$





Chemistry

PART – II

SECTION – A

18. ABC
Sol. Fact.

19. ACD
Sol. Fact.

20. BC
Sol. (A) β -keto acid are easily oxidised by heating.
(B) Gem dioic acid are easily oxidised by heating.

21. B
Sol. Energy involved in formation of 5 mole of $A_2 = 40 \times 5 = 200$ kj
Energy involved information of 3 mole of $A = 50 \times 3 = 150$ kj
Total energy $\Rightarrow$ in decomposition of $A_2B = 200 + 150 = 350$ kj

Energy per mole $= \frac{350}{8} = 43.75$

22. C
Sol. True solution – less than 10^{-9} m
Suspension more than 10^{-7} m
Colloidal between 10^{-9} to 10^{-7} m
Hence suspension > Colloid > True solution.

23. C
Sol. $\lambda = \frac{kn^2}{n^2 - 4}$
Wave number for Balmer transition $n \to 2$

$$\Rightarrow \frac{1}{\lambda} = Rz^2\left[\frac{1}{(2)^2} - \frac{1}{n^2}\right] = R \times (1)^2\left(\frac{1}{(2)^2} - \frac{1}{n^2}\right)$$

$$\lambda = \frac{4n^2}{R\left(n^2 - 4\right)} \Rightarrow K = \frac{4}{R}$$

24. B
Sol. Weight of balloon = 100 Kg = 10^5 g

Volume of balloon = $\frac{4}{3}\pi r^3 = \frac{4}{3} \times \frac{22}{7} \times \left(\frac{21}{2}\right)^3 = 4851 \times 10^3 L$

Weight of gas (He) in balloon = $\frac{PVM}{RT}$

$$\Rightarrow \frac{1 \times 4851 \times 10^3 \times 4}{0.082 \times 300} = 78.87 \times 10^4$$

Total wet of gas + balloon = $78.87 \times 10^4 + 10 \times 10^4 = 88.87 \times 10^4$ g
Weight of air displaced $= 1.2 \times 4851 = 5821.2$ kg pay load = wt. of air displaced – wt. of (balloon + gas) = 4932.42 kg

25. D

Sol. $X \rightleftharpoons Y + Z$

1 0 0

$(1-\alpha)$ α α

$K_c = \frac{\alpha^2}{1-\alpha}$

$\alpha = \sqrt{K_c}$

Similarly others can be calculated.

26. B

Sol. $\pi_1 > \pi_2$ hypertonic

$\pi_1 < \pi_2$ hypotonic

For two solutions,

$\pi_1 = \pi_2$ (isotonic)

27. A

Sol. (P) Concentration of product increase w.r.t time.

(Q) Concentration of reactant decrease w.r.t time

(S) $K = Ae^{-E_a/RT}$

$\ln K = \ln A - \frac{E_a}{RT}$ slope $= -\frac{E_a}{R}$

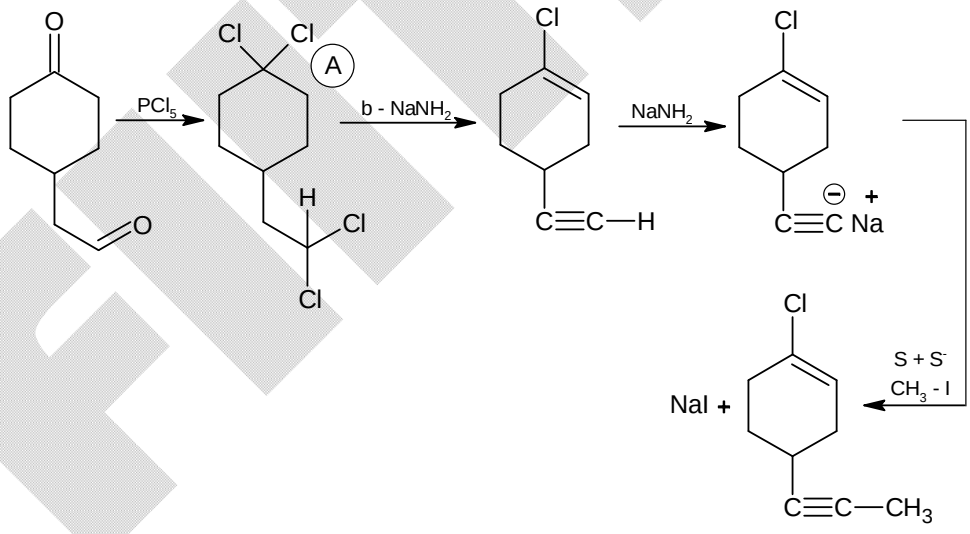
28. B

Sol. In P and Q, elimination takes place, while in case of R and S, substitution takes place.

SECTION – B

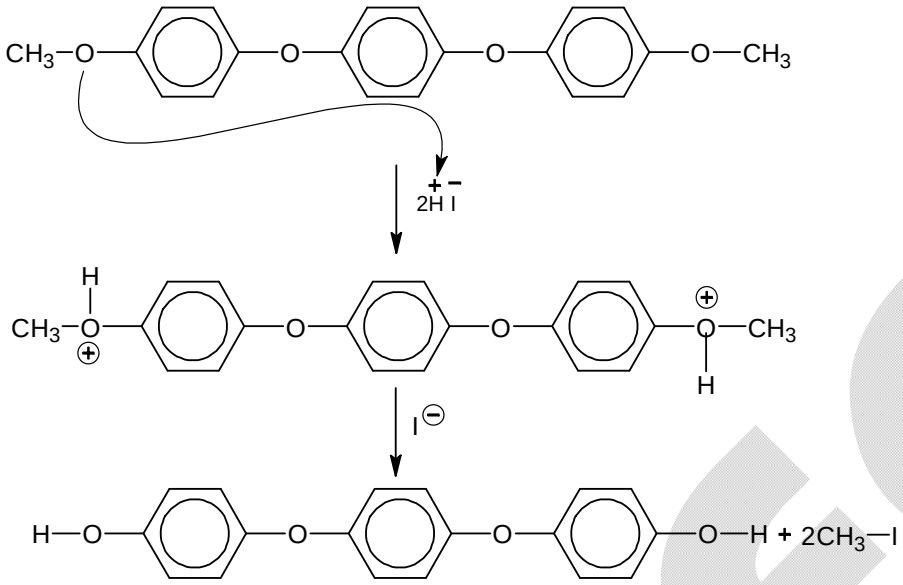
29. 5

Sol.



30. 2

Sol.

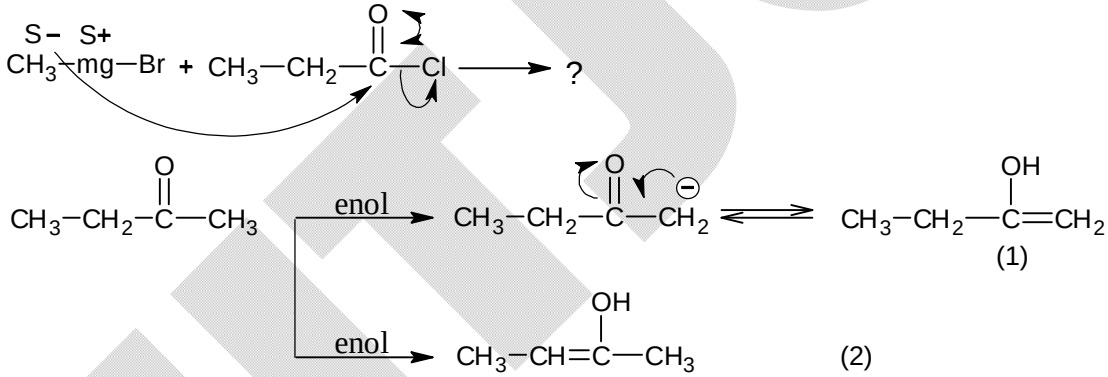


31. 4

Sol. A, B, C, F

32. 3

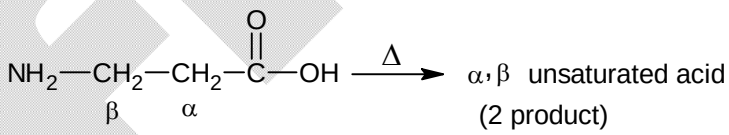
Sol.



Total no. of product are $\Rightarrow$ 3

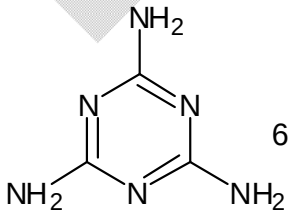
33. 2

Sol.



34. 6

Sol.



6 lone pairs

Mathematics

PART – III

SECTION – A

35. BD

Sol. $P(M)=\frac{10^6}{10^9}$

Event $\frac{N}{M}$ occurs when each of the diagonal elements of A is non-zero

$\therefore P\left(\frac{N}{M}\right)=\frac{P(N\cap M)}{P(M)}=\frac{729}{1000}$

36. BC

Sol. Define domain.

37. BCD

Sol. $\log_3 x\,.\log_x y=\log_3 x\,.\log_y x\,;\quad x\neq 1$

$\Rightarrow x=y$

$\Rightarrow \log_x\left(z-(x-y)^2\right)=4 \quad \Rightarrow z=x^4$

38. B

Sol. $50,a_1,a_2,a_3,.....,a_n,200$ are in A.P $\frac{1}{50},\frac{1}{h_1},\frac{1}{h_2},\frac{1}{h_3},....,\frac{1}{h_n},\frac{1}{200}$ are in A.P $50,\frac{10000}{h_n},\frac{10000}{h_{n-1}},....$

$\frac{10000}{h_1}, 200$ are in A.P and it is identical with series

$50,a_1,a_2,a_3,....,a_n,200$

$\therefore a_2h_{n-1}=10000$

39. B

Sol. $x_1\,1\,x_2\,2\,x_3\,3\,x_4\,4\,x_5$ where x_i denotes number of 7's then, $x_1+x_2+x_3+x_4+x_5=10$

where $x_1,x_5\geq 0$ and $x_2,x_3,x_4\geq 2$

We get $C_4^{4+5-1}\times 4!=14\times 5!$

40. B

Sol. Let $x=a\cos^2\theta+b\sin^2\theta$

$\therefore I=2(b-a)^8\int_0^{\frac{\pi}{2}}\sin^7\theta\cos^9\theta\,d\theta$

$I=\frac{(b-a)^8}{280}$

41. A

Sol. $(\vec{a}\,.\,\vec{x})\vec{b}=\vec{c}+\vec{x}$ taking dot with $\vec{a}$

$\Rightarrow(\vec{a}\,.\,\vec{x})=\frac{(\vec{c}\,.\,\vec{a})}{(\vec{b}\,.\,\vec{a}-1)}$ now substituting it in

Given equation $\vec{x}=\dfrac{\left(\vec{c}\,.\,\vec{a}\right)}{\left(\vec{b}\,.\,\vec{a}-1\right)}\vec{b}-\vec{c}$

42. D

Sol. (P) The abscissa of vertex of $y=x^2+2mx-1$ is $-m$, which must be positive, then m can take values $-34,-1,-17$ and -51.

(Q) $f(7)+f(-7)=-10\Rightarrow f(7)=-17$

$\Rightarrow f(7)+17\sin x=-17+17\sin x$

$\Rightarrow$ which can take value $-34,-1$ and -17

(R) Here $f(x)$ must be a linear function. Let $f(x)=ax+b$

$\Rightarrow f(ax+b)=6x-ax-b$

$\Rightarrow a=2,-3$ and $b=0$

$\therefore f(x)=2x$ or $f(x)=-3x$.

(S) Here $f(x)$ nust be a linear function. Let

$f(x)=ax+b\quad\therefore f(0)=0\therefore b=0$

$f(x^2+1)=f^2(x)+1\therefore f(x)=x$.

43. B

Sol. (P)(Q) draw diagram and find number of points.

(R) $A.M\geq G.M$

$\Rightarrow 2abc=1+a^2b^2c^2\Rightarrow abc=1$

Hence $(a+b+c)_{min}=3$

$(S)[x]=0$

$\therefore I=25\int_0^{\frac{\pi}{4}}(\tan^6x+\tan^4x)dx$

$\Rightarrow I=25\int_0^{\frac{\pi}{4}}\tan^4x\sec^2x\,dx=5$

44. D

Sol. Let end points of major axis are A, B And centre is C. And point of intersection of major axis with directrix is M. Let S(3, 4) and S' is other focus.

Foot of $\perp$ from focus on directrix is

$\Rightarrow\dfrac{x-3}{1}=\dfrac{y-4}{1}=-\dfrac{(3+4-1)}{2}\Rightarrow M(0,1)$

$\therefore A=(2,3)$ and $B=(6,7)$; Centre $C(4,5)$

$ae=CS=\sqrt{2}\quad\therefore a=2\sqrt{2}$ and $b=\sqrt{6}$

45. C

Sol. (P) f(x) is continuous decreasing function

$\Rightarrow\int_1^3f(x)dx+\int_3^1f^{-1}(x)dx$

$I_1=\int_3^1f^{-1}(x)dx$ Let $x=f(t)$

$\int_1^3f(x)dx+I_1=3f(3)-f(1)=0$

(Q) $\left(\frac{5-3\cos 3x}{3+\cos 5x}\right)_{max} = \frac{5+3}{3-1} 4$ at $x = \pi$

(R) $f'(x) = 3x^2 + 2px + q$ has roots 1 and 3

$\Rightarrow p = -6$ and $q = 9$

(S) $\int_0^1 \frac{dx}{(x+1)(x+2)} = \ln\frac{4}{3}$

$\therefore a = 4, b = 3$

SECTION – B

46. 115

Sol. $z = e^{\frac{2n\pi}{100}i}$ where n = 0, 1, 2, ……99

They are located on the vertices of a regular 100 sided polygon, having centre origin.

Let $z_1 = 1$

$\therefore |z_1 + z_2|^2 = \left(1+\cos\frac{2n\pi}{100}\right)^2 + \left(\sin\frac{2n\pi}{100}\right)$

$\Rightarrow |z_1 + z_2|^2 = 2 + 2\cos\frac{2n\pi}{100} \geq 2 + \sqrt{3}$

$-8.33 \leq n \leq 8.33$

$\therefore n(S) = 100 \times 99$

$\Rightarrow n(E) = 100 \times 16$

$\Rightarrow P(E) = \frac{16}{99} = \frac{p}{q} \Rightarrow p + q = 115$

47. 351

Sol. $f(x) = x + \sqrt{1 = x^2}$

$\therefore$ range $= \left[-1, \sqrt{2}\right]$

Number of integers = 3, i.e., -1, 0, 1.

48. 5

Sol. Both the circles are orthogonal to each other and vertices $C(0, 4)$ and $D(3, 0)$ are centres of circles and CD will diameter of circumcircle $= \sqrt{9+16} = 5$.

49. 2

Sol. $\frac{dx}{dy} = \frac{12t^2}{12t^3} = \frac{1}{t}$

$\Rightarrow \frac{d^2x}{dy^2} = -\frac{1}{12t^5}, \therefore$ for constant value n = 5

50. 10

Sol. $\Delta = \frac{1}{2}.8.8 - \frac{2}{3}.32 = \frac{32}{3}$

51. 129

Sol. $\left(a^2\cot 9^0 + d^2\tan 9^0\right) + \left(b^2\cot 27^0 + c^2\tan 27^0\right)$

Given, $a+b+c+d=5$

Let $a + d = k_1$ and $b + c = k_2$

Let $L_1 = a^2\cot 9^0 + d^2\tan 9^0$

$\Rightarrow L_1 = a^2\left(\cot 9^0 + \tan 9^0\right) - ak_1\tan 9^0 + k_1^2\tan 9^0$

This is quadratic in a whose minimum value is obtained at

$a = -\dfrac{B}{2A} = k_1\sin^2 9^0$

Similarly, let $L_2 = b^2\cot 27^0 + c^2\tan 27^0$

$\therefore b = k_2\sin^2 27^0$

$\therefore \left(L_1\right)_{min} = k_1^2\sin 9^0\cos 9^0$

And $\left(L_2\right)_{min} = k_2^2\sin 27^0\cos 27^0$

$\therefore$ minimum value $= \left(L_1\right)_{min} + \left(L_2\right)_{min}$

$\Rightarrow k_1^2\sin 9^0\cos 9^0 + k_2^2\sin 27^0\cos 27^0$

$\Rightarrow \dfrac{1}{2}\left(k_1^2\left(\sin 18^0 + \cos 36^0\right) - 10k_1\cos 36^0 + 25\cos 36^0\right)$

Quadratic in k_1 $\therefore$ minimum value is $-\dfrac{D}{4A}$

$\text{Min} = \dfrac{25}{2}\dfrac{\sin 18^0\cos 36^0}{\sin 18^0 + \cos 36^0} = \dfrac{\sqrt{125}}{4}$